

ProbOS — Month 7 Pre-Plan

Pharma GPU and C++ Parity

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Month goal

Bring `PharmaStabilityModel` to the same GPU and C++ performance-path maturity as `BatteryModel2Cell`.

Week 26 – GPU dispatch for `PharmaStabilityModel`

- CuPy dispatch mirroring the battery model’s `xp` pattern exactly
- Honest benchmark against CPU/C++ – do NOT assume GPU wins

Week 27 – C++ port

- `PharmaCell`-equivalent in C++
- Native GoogleTest coverage from day one
- `pybind11` bindings

Week 28 – Cross-language validation

- Statistical agreement, not bit-exact, given different RNGs

Week 29 – Dashboard integration

Exit criteria

Pharma has full GPU and C++ parity with battery, honestly benchmarked.